

CSE5DMI Data Mining

Week 05
Classification: Alternative Techniques

Objective

- Interpret true-positive and false-positive behaviour rather than relying on accuracy alone.
- Explain why class imbalance changes the usefulness of common performance measures.
- Read an ROC curve and compare classifiers across decision thresholds.
- Describe how a rule-based classifier represents and applies IF–THEN rules.

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Topics

- ROC
- Rule-based classifier

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Review-Week3&4

- Model evaluation

Measure	Question answered	Formula
Accuracy	What proportion of all predictions is correct?	(TP + TN) / (TP + TN + FP + FN)
Precision	When the model predicts positive, how often is it correct?	TP / (TP + FP)
Recall	Of all actual positives, how many are found?	TP / (TP + FN)

- Decision tree
  - We can solve a classification problem by asking a series of carefully crafted questions about the attributes of the test instance.

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Evaluating performance with class imbalance

- Confusion matrix
  - Provides a concise representation of classification performance on a given test dataset
  - True positive (TP), False positive (FP), False negative (FN), True negative (TN)

Simple case study

	Predicted +	Predicted -
Actual +	99	0
Actual -	1	0

Accuracy: 99 Normal, 1 Fraud

Case: The model predicts “all Normal” (100 samples are predicted as normal)

"What's the accuracy?”
Would you trust this model?

Can be misleading

Confusion matrix is important, let's see the other performance measures

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Evaluating performance with class imbalance

Other measures

- Sensitivity (in medical community)
  - Also known as True positive rate(TPR). TPR=TP/(TP+FN)
  - In information retrieval, it is also called recall (r)
- Specificity
  - Also known as True negative rate(TNR). TNR=TN/(FP+TN)
- False positive rate (FPR)
  - FPR=1-TNR
  - FPR=FP/(FP+TN)
- False negative rate (FNR)
  - FNR=1-TPR
  - FNR=FN/(FN+TP)

Case Example

Given Confusion matrix

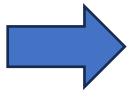
	Predicted +	Predicted -
Actual +	99	0
Actual -	1	0

In the case of Fraud detection, we have Accuracy = 99%
But no fraud is detected.
Need more than Accuracy

TPR (sensitivity/recall) = 99 / (99 + 0) = 1:
the model detects 100% of actual positives.

TNR (specificity) = 0 / (1 + 0) = 0:
The model failed to correctly identify any of the actual negative instances. It predicted every negative instance as positive.

FPR = 1 / (1+0) = 1
How many negatives were wrongly classified as positive?
1 means, **all negative samples were incorrectly classified**.

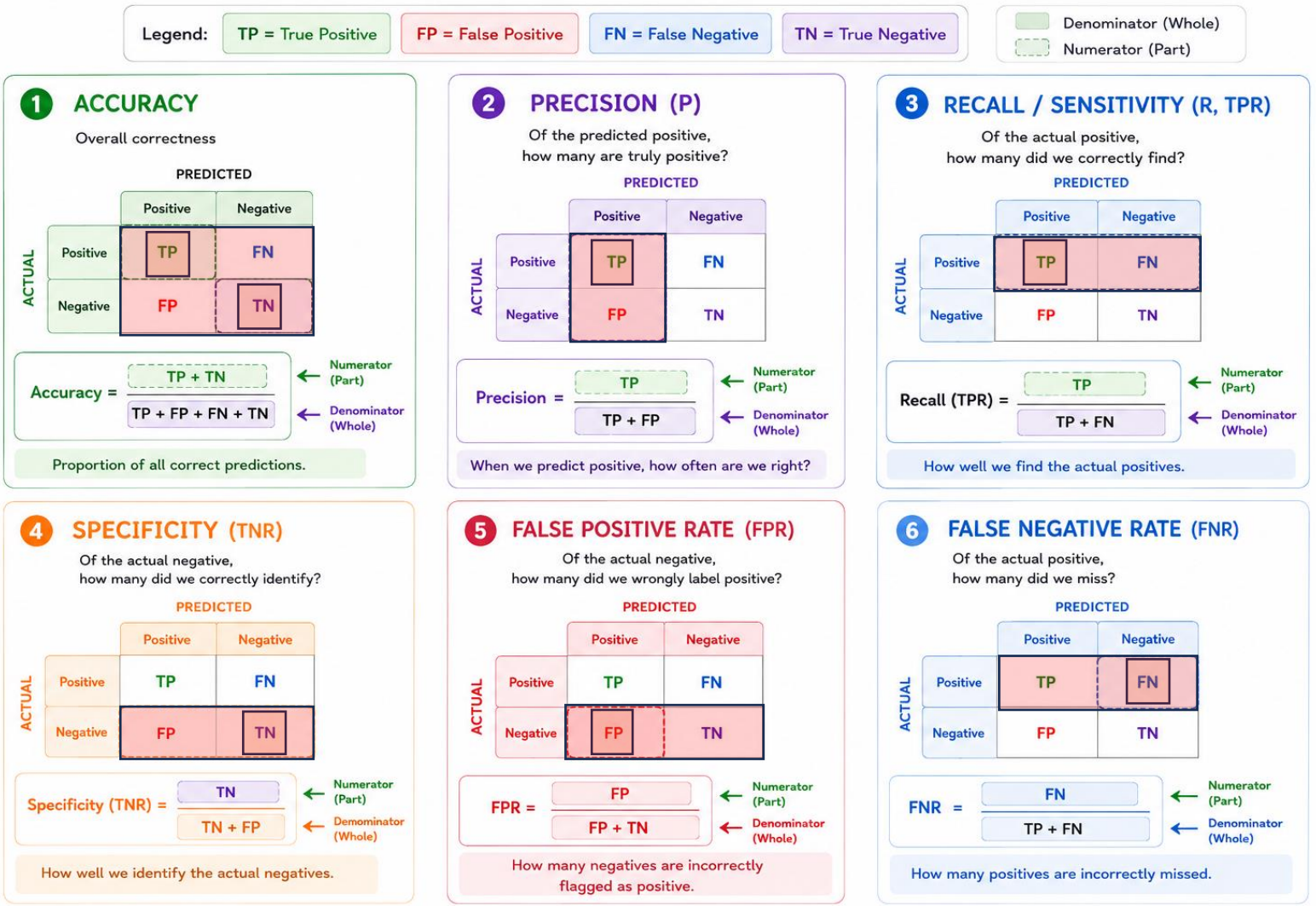


Every negative case became a false alarm.

FNR = 0 / (0+99) = 0
How many positives were wrongly classified as negative?
No positive cases were missed.

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Summary – Core Classification Performance Measures



*Illustrations were generated with the assistance of AI-tool with review and manual correction

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Evaluating performance with class imbalance

- Given
  - The skew among the classes (α). α=P/(P+N)
  - The total number of instances N,
  - Confusion matrix can be calculated using TPR and TNR.

P = number of positive instances
N = number of negative instances

	Predicted +	Predicted -	
Actual +	TPR × α × N	(1-TPR) × α × N	α × N
Actual -	(1-TNR) × (1-α) × N	TNR × (1-α) × N	(1-α) × N
			N

- A evaluation measure that is sensitive to the skew is *precision*.
  - Precision = TP/(TP+FP)
  - Precision is also referred to as positive predicted value (PPV)
  - a useful measure for highly skewed test sets where the positive predictions, even though small in numbers, are required to be mostly correct.

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Example: Computing the Confusion Matrix from TPR, TNR, and Class Skew

Scenario: N = 1,000;

α = 200/1000 = 0.20;

TPR = 90%;

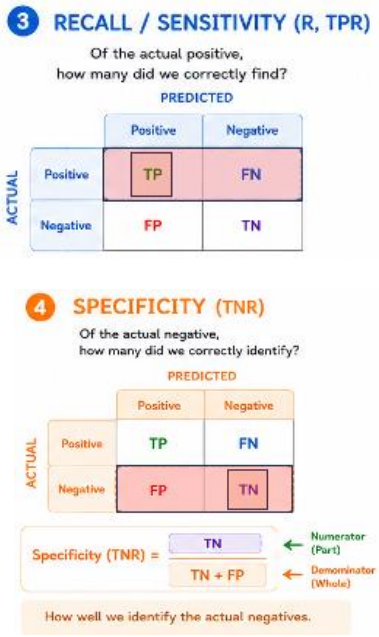
TNR = 80%

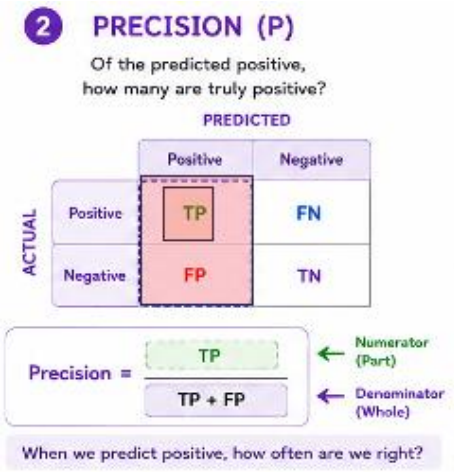
Actual positives =

$\alpha \times N = 0.20 \times 1{,}000 = 200$

Actual negatives =

$(1 - \alpha) \times N = 0.80 \times 1{,}000 = 800$





	Predicted +	Predicted –	
Actual +	TP = 180	FN = 20	200
Actual –	FP = 160	TN = 640	800
Total	340	660	1,000

Precision
TP / (TP + FP)
= 180 / (180 + 160)
= 180 / 340
= 52.9%
Interpretation
High TPR does not guarantee high precision when many false positives exist.

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F-measure - with Example

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad \text{Precision} = \frac{TP}{TP + FP} = \frac{40}{40 + 10} = 0.8$$

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	40	10
	Class=No	10	40

Precision (p) = 0.8

TPR = Recall (r) = 0.8

FPR = 0.2

F - measure (F) = 0.8

Accuracy = 0.8

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{40}{40 + 10} = 0.8$$

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	40	10
	Class=No	1000	4000

Precision (p) =~ 0.04

TPR = Recall (r) = 0.8

FPR = 0.2

F - measure (F) =~ 0.08

Accuracy =~ 0.8

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Evaluating performance with class imbalance

- A measure that is closely related to precision is the false discovery rate (FDR).
  - FDR=1-precision
  - FDR=FP/(TP+FP)

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Summary: Measures of Classification Performance

	PREDICTED CLASS		
ACTUAL CLASS		Yes	No
	Yes	TP	FN
	No	FP	TN

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall (TPR)} = \frac{TP}{TP + FN}$$

$$\text{Specificity (TNR)} = \frac{TN}{TN + FP}$$

$$\text{FPR} = \frac{FP}{FP + TN} = 1 - \text{Specificity}$$

$$\text{FNR} = \frac{FN}{FN + TP} = 1 - \text{Recall}$$

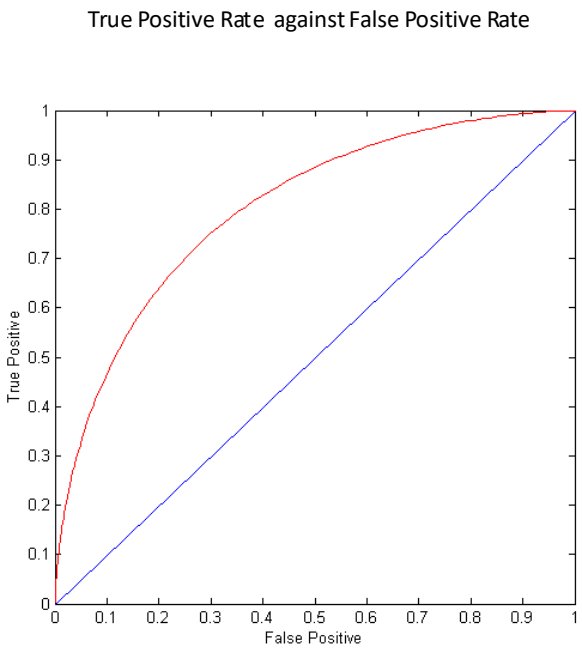
$$\boxed{\text{FDR} = 1 - \text{Precision} = \frac{FP}{TP + FP}}$$

$$\boxed{F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}}$$

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ROC (Receiver Operating Characteristic)

- Trade-off between detection rate (TPR) and false alarm rate (FPR)
  - Performance of a model represented as a point in an ROC curve
  - Plots TPR against FPR
- Developed in 1950s for signal detection theory to analyze noisy signals
- To plot, the classifier must produce continuous-valued output
  - Outputs are used to rank test records, from the most likely positive class record to the least likely positive class record
  - Many classifiers produce only discrete outputs (i.e., predicted class)
    - How to get continuous-valued outputs?



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ROC Curve

(TPR,FPR) (see example):

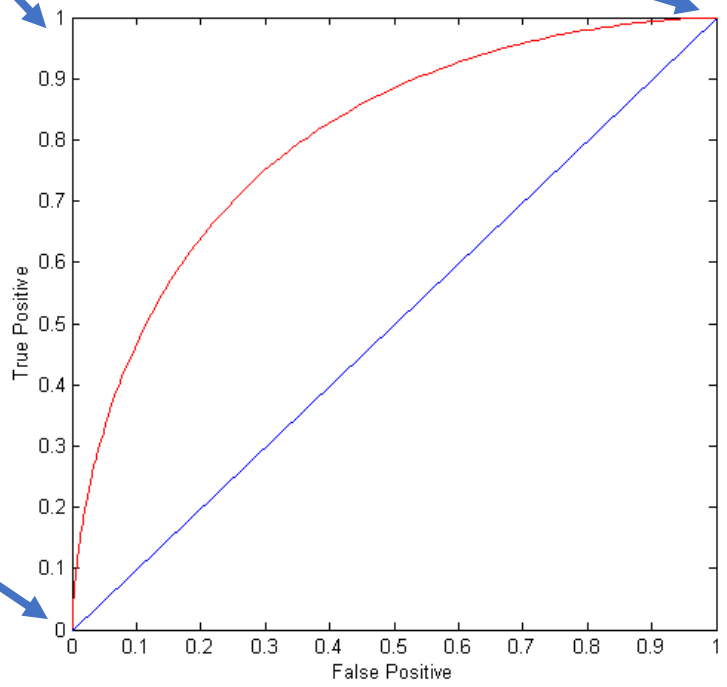
- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive
- (1,0): ideal

- Diagonal line:
  - Random guessing
  - Below diagonal line:
    - prediction is opposite of the true class
    - (TP+TN)/(FP+FN) < 1

	Predicted +	Predicted –	
Actual +	TP = 600	FN = 0	600
Actual –	FP = 0	TN = 400	400
Total	600	400	1000
TPR = 1 FPR = 0			

	Predicted +	Predicted –	
Actual +	TP = 600	FN = 0	600
Actual –	FP = 400	TN = 0	400
Total	1000	0	1000
TPR = 1 FPR = 1			

	Predicted +	Predicted –	
Actual +	TP = 0	FN = 600	600
Actual –	FP = 0	TN = 400	400
Total	0	1000	1000
TPR = 0 FPR = 0			



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How to construct an ROC curve

Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

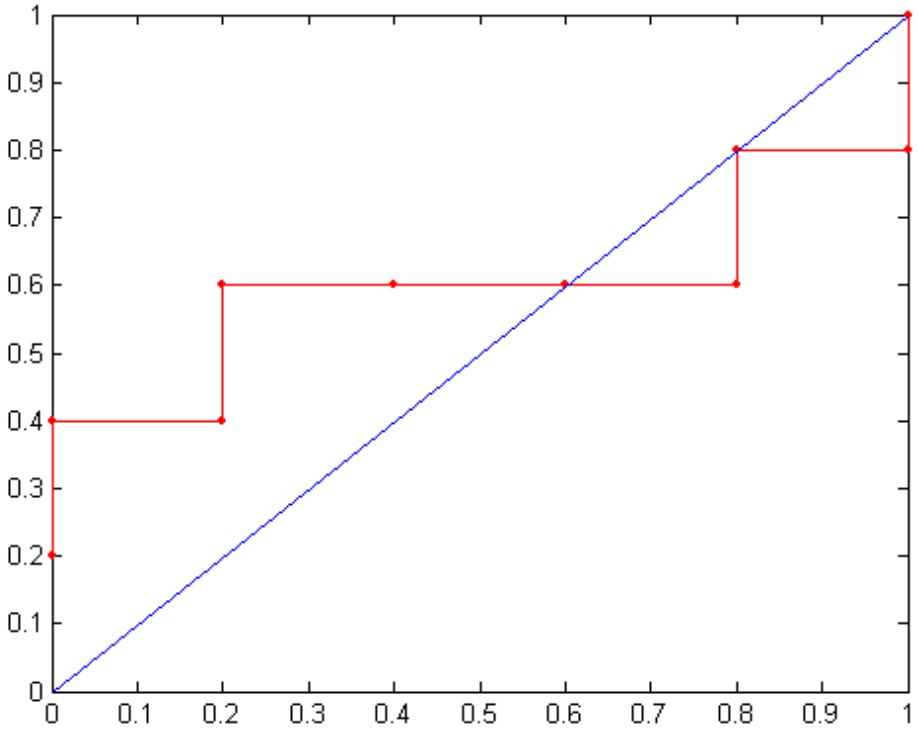
- Use a classifier that produces a continuous-valued score for each instance
  - The more likely it is for the instance to be in the + class, the higher the score
- Sort the instances in decreasing order according to the score
- Apply a threshold at each unique value of the score
- Count the number of TP, FP, TN, FN at each threshold
  - TPR = TP/(TP+FN)
  - FPR = FP/(FP + TN)

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How to construct an ROC curve

	Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=		0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
	TP	5	4	4	3	3	3	3	2	2	1	0
	FP	5	5	4	4	3	2	1	1	0	0	0
	TN	0	0	1	1	2	3	4	4	5	5	5
	FN	0	1	1	2	2	2	2	3	3	4	5
→	TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→	FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

ROC Curve:



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How to construct an ROC curve : Demo

Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=	0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
TP	5	4	4	3	3	3	3	2	2	1	0
FP	5	5	4	4	3	2	1	1	0	0	0
TN	0	0	1	1	2	3	4	4	5	5	5
FN	0	1	1	2	2	2	2	3	3	4	5
TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- TPR=TP/(TP+FN)
- FPR=FP/(FP+TN)

TP = 1	FN = 4
FP = 0	TN = 5

TP

FP

FN

TN

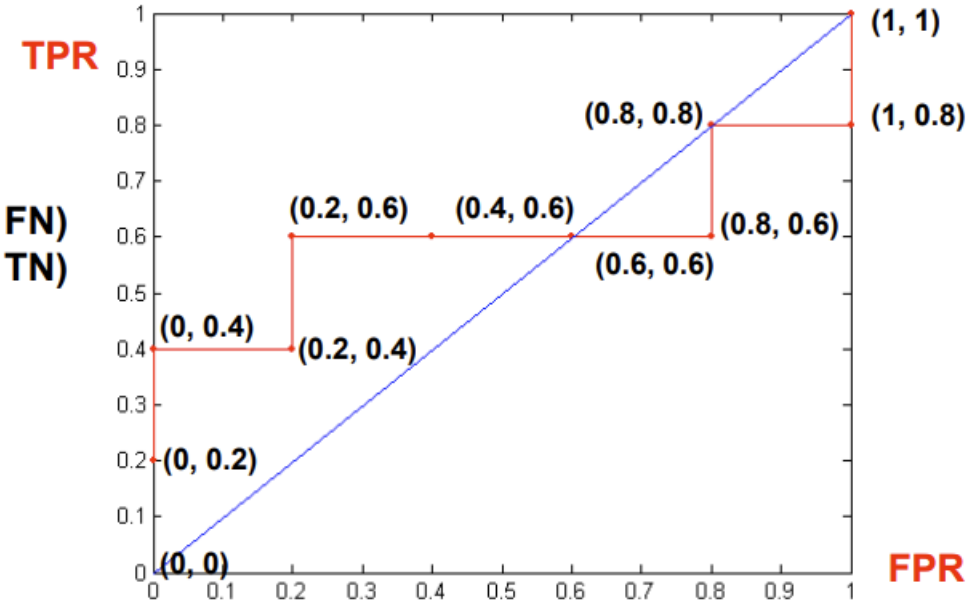
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How to construct an ROC curve

	Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=		0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
	TP	5	4	4	3	3	3	3	2	2	1	0
	FP	5	5	4	4	3	2	1	1	0	0	0
	TN	0	0	1	1	2	3	4	4	5	5	5
	FN	0	1	1	2	2	2	2	3	3	4	5
→	TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→	FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

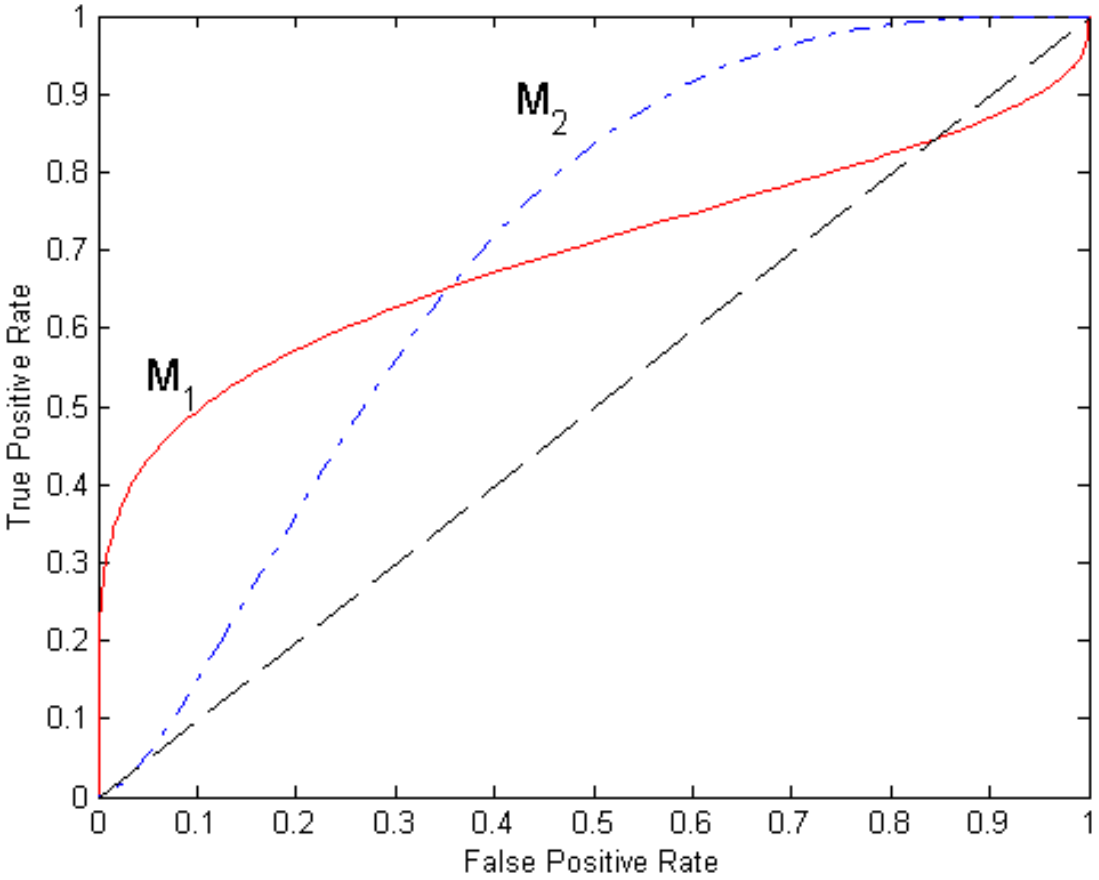
ROC Curve:

TP rate, TPR = TP/(TP + FN)
FP rate, FPR = FP/(FP + TN)



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Using ROC for Model Comparison



- No model consistently outperforms the other
  - M_1 is better for small FPR
  - M_2 is better for large FPR
- Area Under the ROC curve (AUC)
  - Ideal:
    - Area = 1
  - Random guess:
    - Area = 0.5

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Rule-Based Classifier

Classify records by using a collection of "if…then…" rules

Rule: (***Condition***) $\rightarrow$ ***y***

- where
  - *Condition* is a conjunction of tests on attributes
  - *y* is the class label
  - *Left hand side (LHS): rule antecedent or condition*
  - *Right hand side (RHS): rule consequent*
- Examples of classification rules:
  - (Blood Type=Warm) $\wedge$ (Lay Eggs=Yes) $\rightarrow$ Birds
  - (Taxable Income < 50K) $\wedge$ (Refund=Yes) $\rightarrow$ Evade=No

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Rule-based Classifier (Example)

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
human	warm	yes	no	no	mammals
python	cold	no	no	no	reptiles
salmon	cold	no	no	yes	fishes
whale	warm	yes	no	yes	mammals
frog	cold	no	no	sometimes	amphibians
komodo	cold	no	no	no	reptiles
bat	warm	yes	yes	no	mammals
pigeon	warm	no	yes	no	birds
cat	warm	yes	no	no	mammals
leopard shark	cold	yes	no	yes	fishes
turtle	cold	no	no	sometimes	reptiles
penguin	warm	no	no	sometimes	birds
porcupine	warm	yes	no	no	mammals
eel	cold	no	no	yes	fishes
salamander	cold	no	no	sometimes	amphibians
gila monster	cold	no	no	no	reptiles
platypus	warm	no	no	no	mammals
owl	warm	no	yes	no	birds
dolphin	warm	yes	no	yes	mammals
eagle	warm	no	yes	no	birds

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

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Application of Rule-Based Classifier

A rule *r* covers an instance **x** if the attributes of the instance satisfy the condition of the rule

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
hawk	warm	no	yes	no	?
grizzly bear	warm	yes	no	no	?

The rule R1 covers a hawk => Birds

The rule R3 covers the grizzly bear => Mammals

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When building a rule-based classifier, we can define many different combinations of rules. However, these rule sets may differ in accuracy, coverage, complexity, and how well they generalise to new data.



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We need evaluation measures to determine which rule set is better.



Aim:

Determine the best rule set

Rule Coverage and Accuracy

Example

1) This slide evaluates the classification rule:

$$(\text{Marital Status} = \text{Single}) \rightarrow \text{Class} = \text{No}$$

2) Coverage of a rule – Coverage(r)

$$= \frac{\text{records satisfying the antecedent}}{\text{all records}}$$

3) Accuracy

Among the records that satisfy the rule condition, what percentage have **Class = No**?

Given a dataset D and classification rule r: A->y,

Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(Status=Single) → No

Coverage = 40%, Accuracy = 50%

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Rule Coverage and Accuracy

Given a dataset D and classification rule r: A->y,

$$\text{Coverage}(r) = \frac{|A|}{|D|}$$

$$\text{Accuracy}(r) = \frac{|A \cap y|}{|A|},$$

Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(Status=Single) → No

Coverage = 40%, Accuracy = 50%

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How does Rule-based Classifier Work?

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

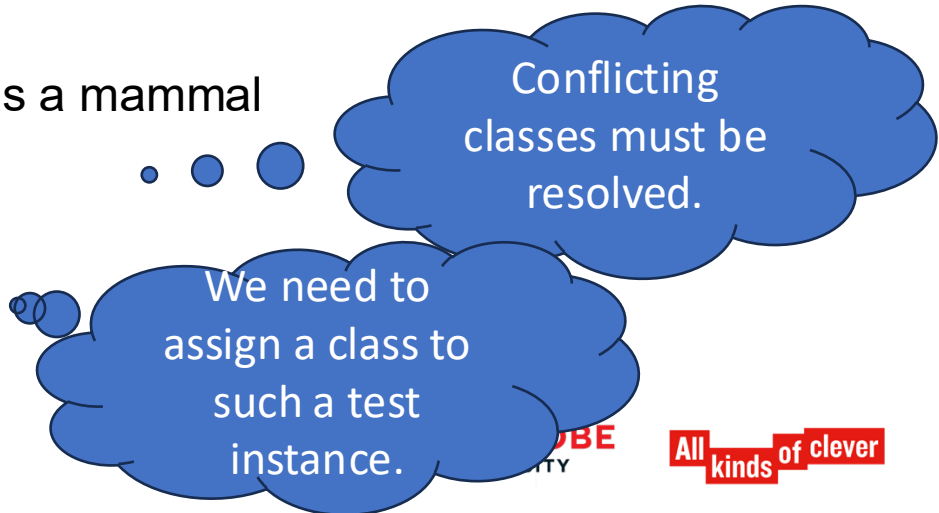
R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
lemur	warm	yes	no	no	?
turtle	cold	no	no	sometimes	?
dogfish shark	cold	yes	no	yes	?

A lemur triggers rule R3, so it is classified as a mammal

A turtle triggers both R4 and R5

A dogfish shark triggers none of the rules



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How to design rules?

- we want the rules to cover every instance in the dataset
- We also expect that every instance should not be covered by two rules, or three rules, or multiple rules.



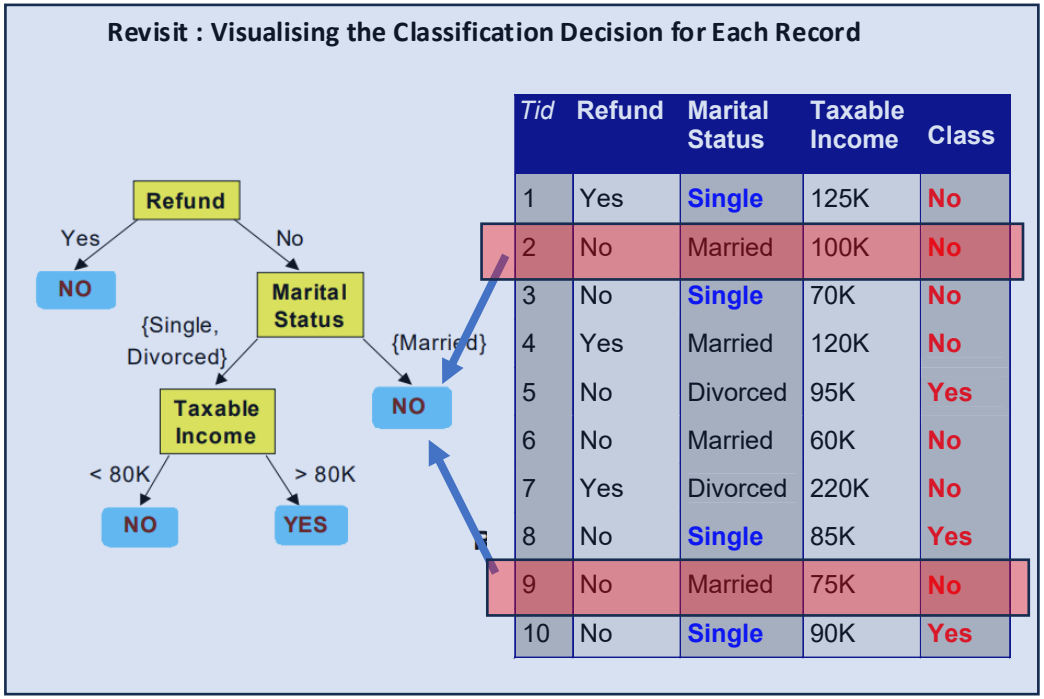
Characteristics of Rule-based Classifier

Mutually exclusive rules

- Classifier contains mutually exclusive rules if the rules are independent of each other
- Every record is covered by at most one rule

Exhaustive rules

- Classifier has exhaustive coverage if it accounts for every possible combination of attribute values - leave no record unclassified.



Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
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10	No	Single	90K	Yes

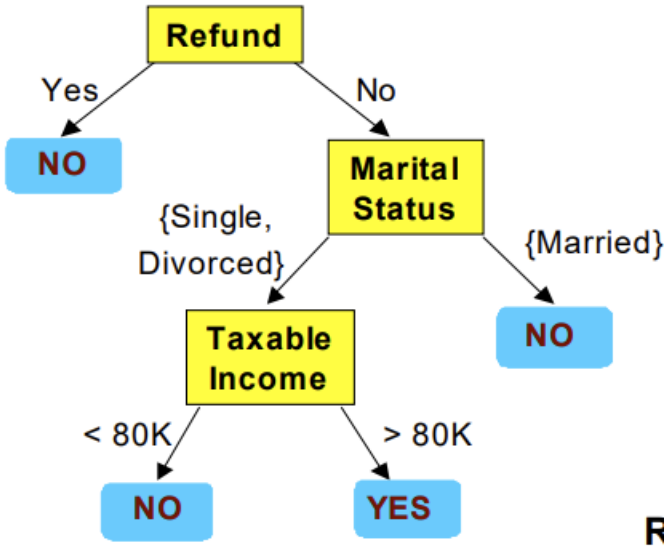
Rule-based classifier

Every record is covered by **exactly one** rule.

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From Decision Trees to Rules

That mutually exclusive and Exhaustive



Classification Rules

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced}, Taxable Income<80K) ==> No

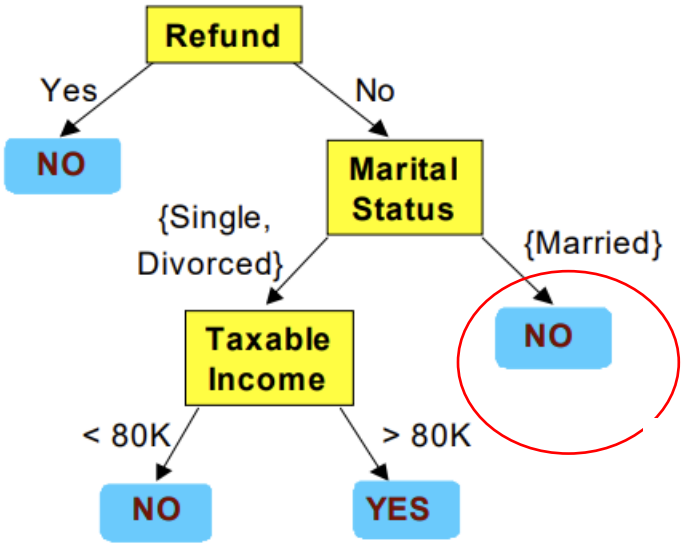
(Refund=No, Marital Status={Single,Divorced}, Taxable Income>80K) ==> Yes

(Refund=No, Marital Status={Married}) ==> No

- Rules are mutually exclusive
- Rule set contains as much information as the tree

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Rules can be simplified



Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Initial Rule: $(\text{Refund=No}) \wedge (\text{Status=Married}) \rightarrow \text{No}$

Simplified Rule: $(\text{Status=Married}) \rightarrow \text{No}$

- Just like decision trees, classification rules can be **pruned or simplified**
  - reduce complexity while maintaining similar predictive performance.
- Conditions that provide little or no additional discriminatory information
  - such as minimal reductions in Gini impurity or entropy

Example

- *The Refund = No condition is redundant: all married records are classified as Cheat = No.*
- *Removing it does not change the predictions on this dataset.*

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Effect of rule simplification

Rules are not mutually exclusive

- A record may trigger more than one rule
- Solution?
  - Ordered rule set
  - Unordered rule set – use voting schemes

Rules are not exhaustive

- A record may not trigger any rules
- Solution?
  - Use a default class

Voting example:

- R1 → **Yes**
- R2 → **No**
- R3 → **Yes**
- R4 → **Yes**

The votes are:

- Yes: 3 votes
- No: 1 vote

Therefore, the final prediction is **Yes**.

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Ordered Rule Set

Rules are rank ordered according to their priority

- An ordered rule set is known as a decision list

When a test record is presented to the classifier

- It is assigned to the class label of the highest ranked rule it has triggered
- If none of the rules fired, it is assigned to the default class

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds
R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes
R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals
R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles
R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
turtle	cold	no	no	sometimes	?

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How to order rules?



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Examples of Ordering

Rule-based ordering

e.g. accuracy

- Individual rules are ranked based on their quality

Class-based ordering

- Rules that belong to the same class appear together

e.g. The class that has the smallest description length is given the highest priority because it is expected to contain the best set of rules.

Rule-based Ordering

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced}, Taxable Income<80K) ==> No

(Refund=No, Marital Status={Single,Divorced}, Taxable Income>80K) ==> Yes

(Refund=No, Marital Status={Married}) ==> No

Class-based Ordering

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced}, Taxable Income<80K) ==> No

(Refund=No, Marital Status={Married}) ==> No

(Refund=No, Marital Status={Single,Divorced}, Taxable Income>80K) ==> Yes

and more...

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Building Classification Rules

Direct Method:

- Extract rules directly from data
- Examples: RIPPER, CN2, Holte's 1R

Indirect Method:

- Extract rules from other classification models (e.g. decision trees, neural networks, etc).
- Examples: C4.5rules

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RIPPER: Sequential Covering

RIPPER:

- A widely used rule induction algorithm.
- It creates rules in the form **IF condition, THEN class**, and repeatedly improves and prunes them to keep the rule set accurate but simple.
- It also works well with noisy data because it uses a validation set to prevent model overfitting.

Procedure

1. Start from an empty rule
2. Grow a rule using the Learn-One-Rule function
3. Remove training records covered by the rule
4. Repeat Step (2) and (3) until stopping criterion is met

Simple example

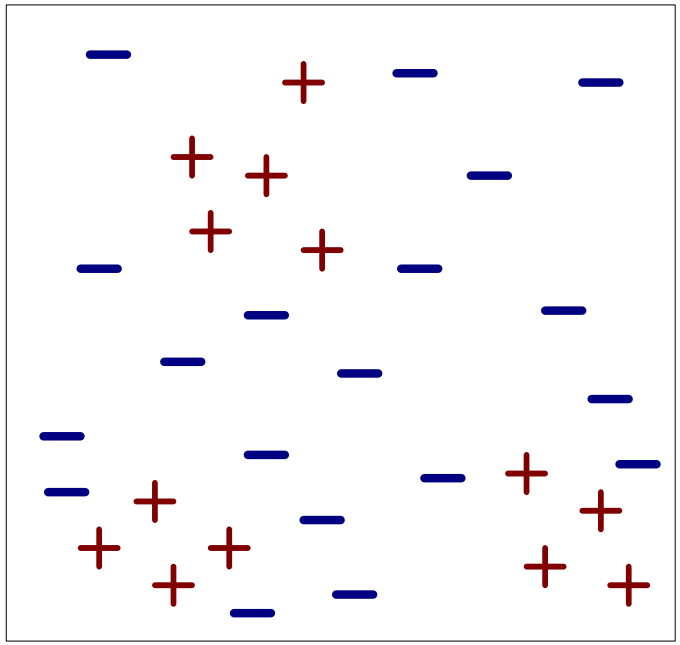
Suppose fraud cases are rare. RIPPER may learn:
IF Amount> 5000 and Country differs from usual country → Fraud

It then checks whether every condition improves performance on validation data.

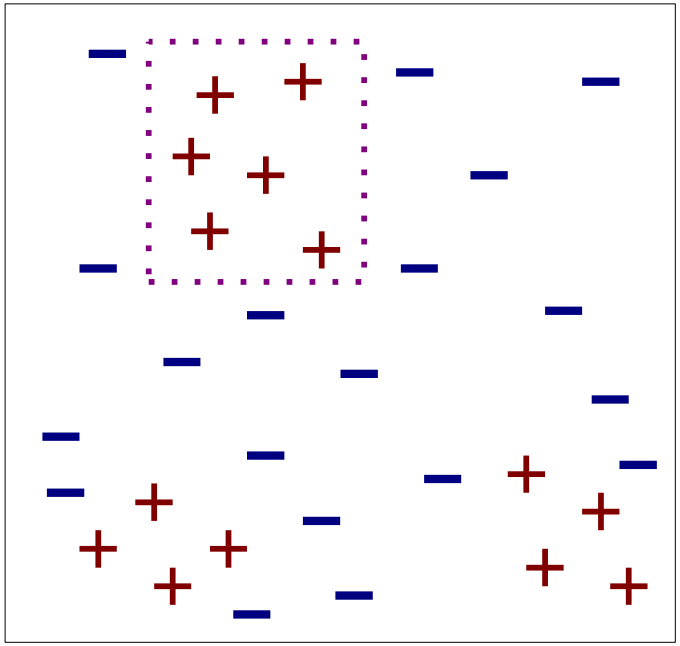
If Amount > $5,000 does not help on unseen examples, RIPPER may remove it:

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Example of Sequential Covering



(i) Original Data

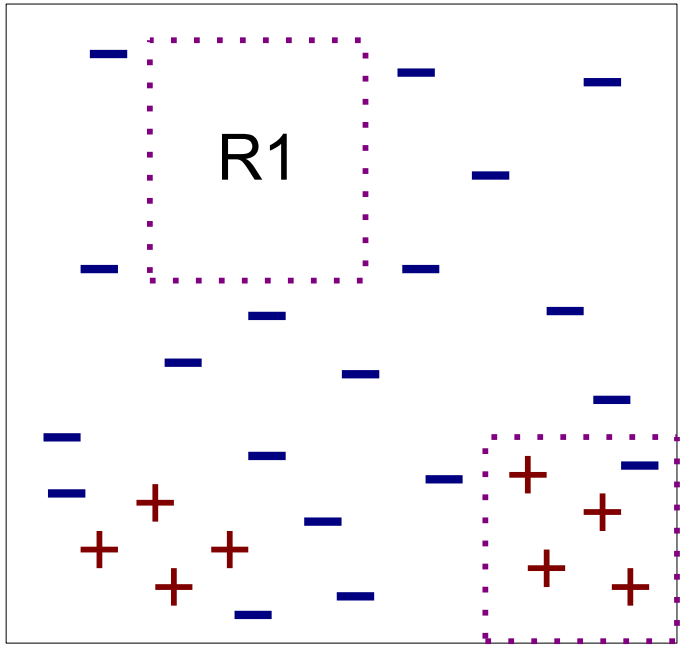


(ii) Step 1

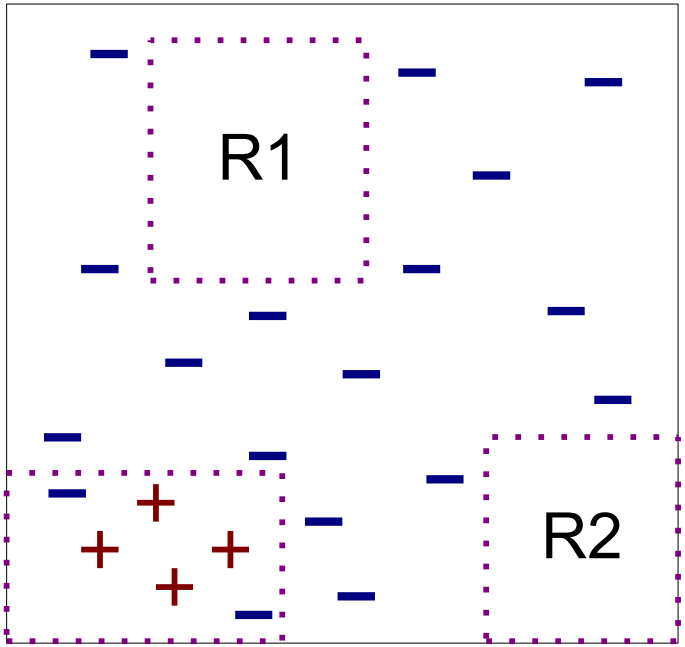
We firstly extract the best rule R1, because the rectangle covers the largest fraction of positive examples.

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Example of Sequential Covering…



(iii) Step 2



(iv) Step 3

Remove training instances covered by R1

Look for the next best Rule, e.g. R2. Repeat the steps

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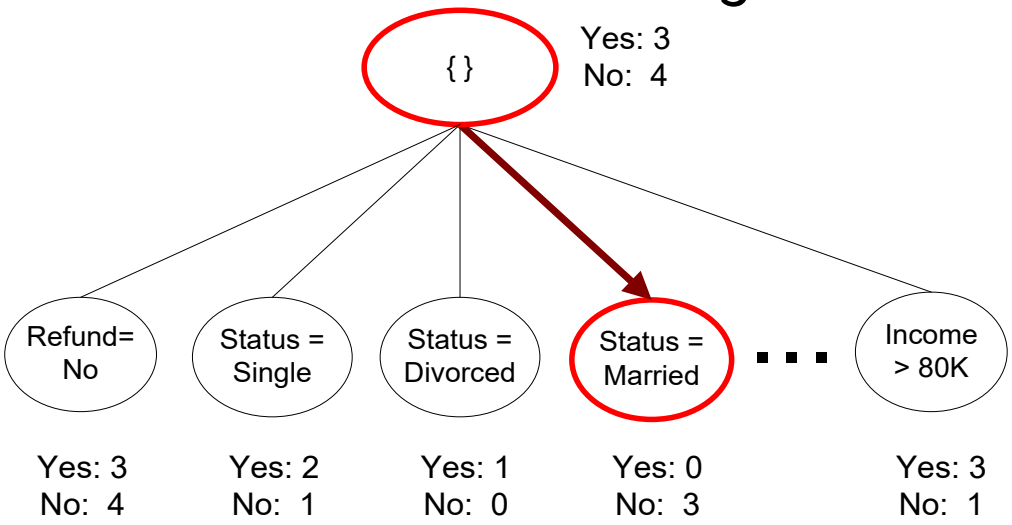
Aspects of Sequential Covering

- Rule Growing
- Instance Elimination
- Rule Evaluation
- Stopping Criterion
- Rule Pruning

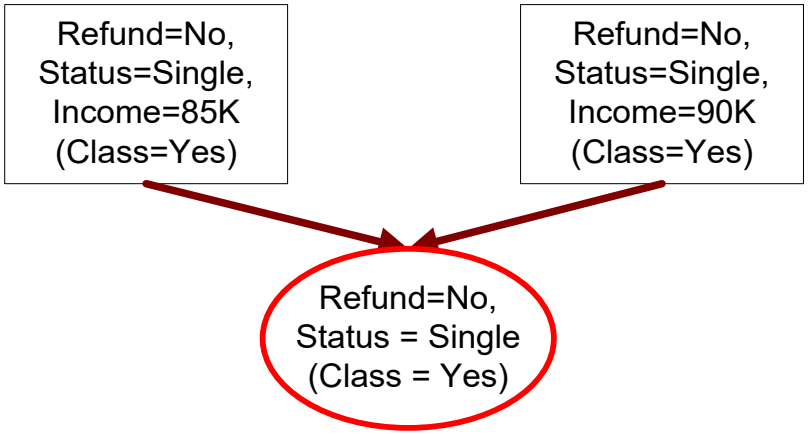
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Rule Growing

- Two common strategies



(a) General-to-specific



(b) Specific-to-general

- In the beginning, empty rule and it covers all training records.
- Once we add one rule from {} to {status-Married), leaves 0 positive records and 3 negative records
- It is useful when constructing a rule for the **No** class, but not for the **Yes** class.

- Remove unnecessary details to create a more general rule
- Prune unnecessary conditions "Income" using **validation data**

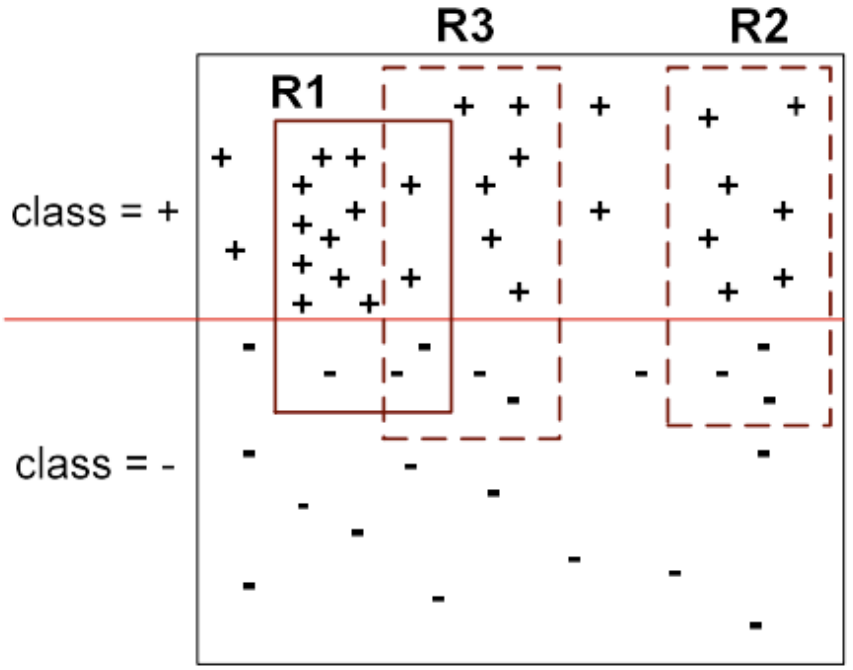
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Positive and Negative Instances Covered by Rules

The rectangular region for R1,R2, and R3 represent the records satisfying the conditions of a learned rule.

For example, R1
Every point inside the R1 region satisfies its conditions.

- The region contains mostly positive examples (correctly cover the right class)
- Some of them negatives (incorrectly covered)

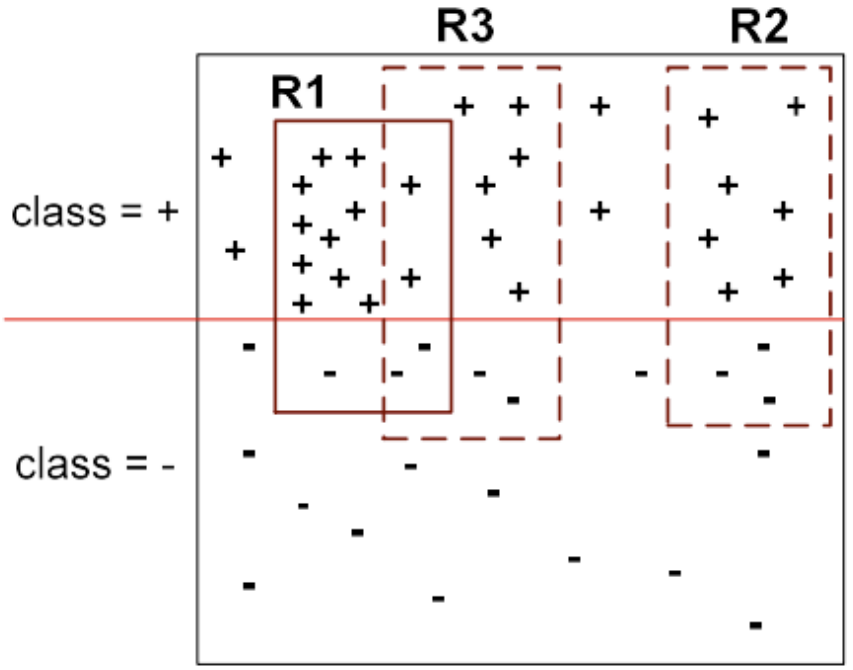


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Instance Elimination

Why do we need to remove instances from a learned rule?

- Next rule will not learn identical dataset that has been learned previously by other rules.
- Prevents repeated coverage of rules and
- Helps each new rule focus on the portion of the dataset not yet handled.



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Example

Assume: Training set: 29 Positive & 21 negative samples

Accuracy of Rules:

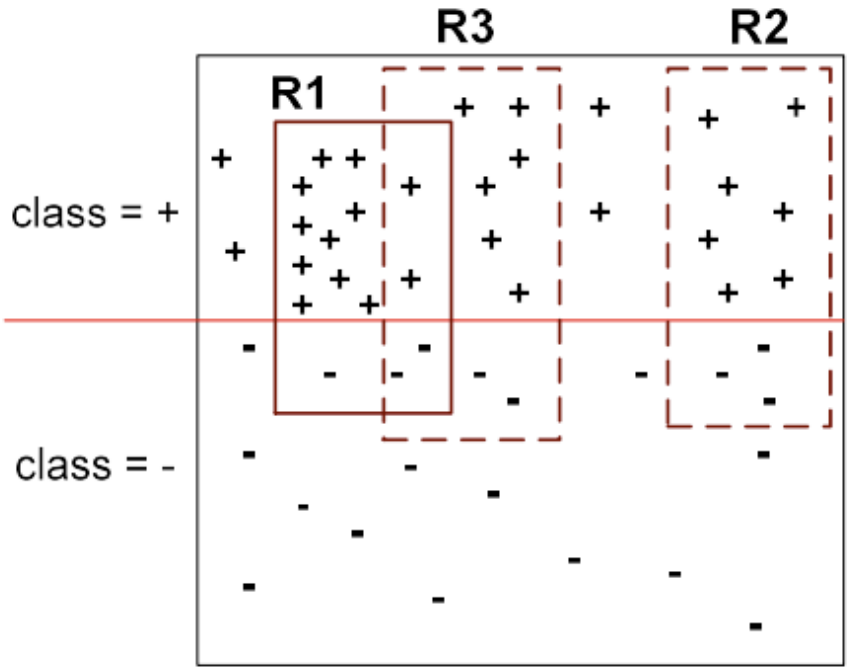
R1 = 12/15 (80%)
R2= 7/10 (70%)
R3 = 8/12 (66.7%)

Next step:

(1) Learn R1 first.
R1 has the highest accuracy:
R1 accuracy = 12/15 = 80%
Initially, R2 appears better than R3:
R2 accuracy = 7/10 = 70%
R3 accuracy = 8/12 = 66.7%

(2) Check the overlap
R2 is disjoint from R1, meaning that R1 and R2 do not cover any of the same records.
R3 overlaps with R1. Some positive and negative records are covered by both rules.

(3) Remove the samples (positive and negative) covered by R1
If positive samples covered by R1 are not removed, the effective accuracy of R3 may be **overestimated**.
If negative samples covered by R1 are not removed, the accuracy of R3 may be **underestimated**.



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Example

Assume: Training set: 29 Positive & 21 negative samples

Accuracy of Rules:

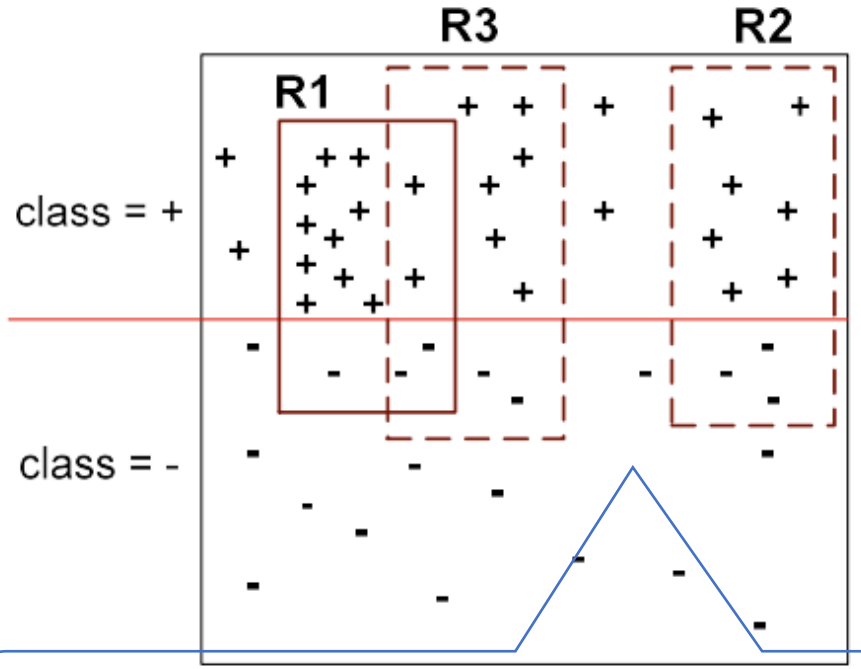
R1 = 12/15 (80%)
R2= 7/10 (70%)
R3 = 8/12 (66.7%)

Next step:

Effect of R3 performance in different

Removal after R1	Remaining records covered by R3	Accuracy of R3
1. Nothing removed	8+, 4-	8/12=66.7%
2. Only R1 positives removed	6+, 4-	6/10=60% (underestimates)
3. Only R1 negatives removed	8+, 2-	8/10=80% (overestimates)
4. Both R1 positives and negatives removed	6+, 2-	6/8=75%

(6) Conclusion
We need to remove both + and – samples covered by R1



(4) Re-evaluate R3
After removing both the positive and negative records covered by R1:
R3 accuracy = 6/8 = 75%
R3 now performs better than R2, which has an accuracy of 70%. Therefore, R3 should be selected.

(5) Compare the resulting rule sets
R1 + R3:
Overall accuracy = 18/23 = 78.3%
R1 + R2:
Overall accuracy = 19/25 = 76%

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Rule Evaluation

FOIL: First Order Inductive Learner – an early rule-based learning algorithm

- **FOIL Information Gain**
- Start with a simple rule (R0).
- Try adding one condition to create R1.
- Count
  - **p** = positive examples covered.
  - **n** = negative examples covered.
- A good condition:
  - keeps many positive examples,
  - removes many negative examples.

$$Gain(R_0, R_1) = p_1 \times \left[log_2 \left(\frac{p_1}{p_1 + n_1} \right) - log_2 \left(\frac{p_0}{p_0 + n_0} \right) \right]$$

- p_0 :number of positive instances covered by R0
- n_0 :number of negative instances covered by R0
- p_1 :number of positive instances covered by R1
- n_1 :number of negative instances covered by R1

- Calculate Information Gain for each candidate condition.
- Choose the condition with the **highest Information Gain** and continue growing the rule.

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Example: Evaluating One Candidate Condition (Income)

Step 1: Initial Rule (R_o)

```
IF {}
THEN Approved = Yes
```

This rule covers **all applicants**.

Covered by R_o	Count
Positive (Approved = Yes)	**6**
Negative (Approved = No)	**4**

Therefore

- $p_o = 6$
- $n_o = 4$

ID	Income	Approved?	Covered by R_o
1	$75K	Yes	✓
2	$82K	Yes	✓
3	$68K	Yes	✓
4	$65K	Yes	✓
5	$70K	Yes	✓
6	$45K	Yes	✓
7	$90K	No	✓
8	$55K	No	✓
9	$40K	No	✓
10	$35K	No	✓

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Example: Evaluating One Candidate Condition

Step 2: Add One Candidate Condition

Try

```
IF Income > $60,000
THEN Approved = Yes
```

Now only applicants with income above $60,000 are covered.

Covered by R_1	Count
Positive	**5**
Negative	**1**

Therefore

- $p_1 = 5$
- $n_1 = 1$

ID	Income	Approved?	Covered by R1
1	$75K	Yes	✓
2	$82K	Yes	✓
3	$68K	Yes	✓
4	$65K	Yes	✓
5	$70K	Yes	✓
6	$45K	Yes	X
7	$90K	No	✓
8	$55K	No	X
9	$40K	No	X
10	$35K	No	X

R_1 covers only the samples that satisfy the rule (Income > $60K) (Sample 1,2,3,4,5,7). Among these covered samples, there are 5 positive examples (p_1) and 1 negative example (n_1).

Now, we can calculate Information Gain

$$Gain(R_0, R_1)$$
$$= p_1 \times \left[log_2\left(\frac{p_1}{p_1 + n_1}\right) - log_2\left(\frac{p_0}{p_0 + n_0}\right) \right]$$
$$= 5 \times \left[log_2\left(\frac{5}{5+1}\right) - log_2\left(\frac{6}{6+4}\right) \right] = 2.37$$

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Extending Rule from R0 -> R1

- How to evaluate?
  - Now FOIL evaluates another condition to further improve the rule.
  - Example

Candidate Rule	p	n	Gain
IF Income > $60K **AND Credit Score > 700**	5	0	**Highest**
IF Income > $60K **AND Age > 30**	4	1	Lower
IF Income > $60K **AND Owns House = Yes**	3	1	Lower

- RIPPER algorithm chooses the conjunct with the *highest* information gain to extend the rule.
  - The process continues until adding new conjuncts no longer improves the information gain measure.

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Stopping criterion and rule pruning

Stopping criterion

- Compute the gain
- If gain is not significant, discard the new rule

Pruning

- Similar to post-pruning of decision trees
- Reduced Error Pruning:
    - Remove one of the conjuncts in the rule
    - Compare error rate on validation set before and after pruning
    - If error improves, prune the conjunct

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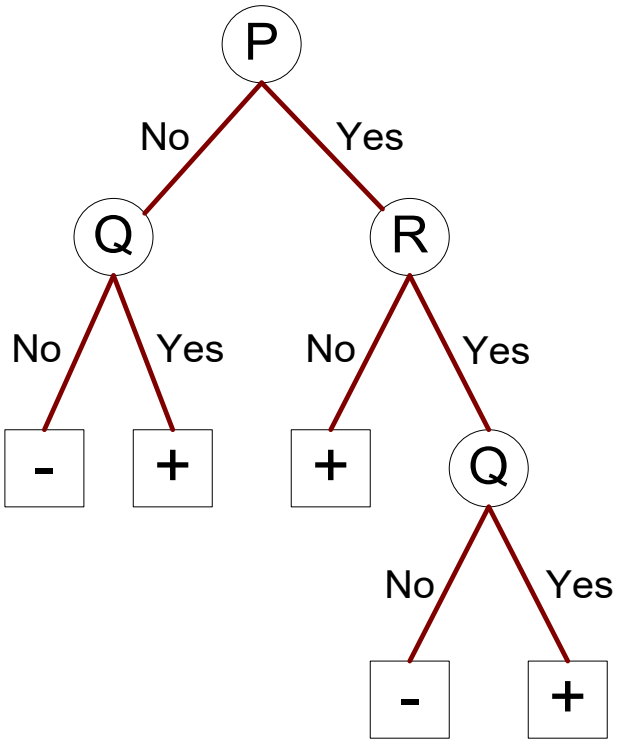
Summary of Direct Method

- Grow a single rule
- Remove Instances from rule
- Prune the rule (if necessary)
- Add rule to Current Rule Set
- Repeat

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Indirect method

In principle, every path from the root node to the leaf node of a decision tree can be expressed as a classification rule.



Rule Set

r1: (P=No,Q=No) ==> -
r2: (P=No,Q=Yes) ==> +
r3: (P=Yes,R=No) ==> +
r4: (P=Yes,R=Yes,Q=No) ==> -
r5: (P=Yes,R=Yes,Q=Yes) ==> +

The rule set always predicts a positive class when the value of Q is Yes.

We may simplify the rules as:
R2 (new): (Q=Yes) == >+

Advantages of Rule-Based Classifiers

- Has characteristics quite similar to decision trees
  - As highly expressive as decision trees
  - Easy to interpret (if rules are ordered by class)
  - Performance comparable to decision trees
    - Can handle redundant and irrelevant attributes
- Better suited for handling imbalanced classes
- Harder to handle missing values in the test set

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Summary

- ROC
- Rule-based classifier
  - Basics
  - Sequential covering

- Assignment 1

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Thank you

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